

Pneumonia Classification from Chest X-Rays Using CNNs

Historical project report - English translation

This report preserves the original project version and its recorded results. For the current model, audited splits, and external evaluations, see README.md and docs/project_report.md. Historical scores below do not describe the current model.

Abstract

This project developed a deep learning system using CNNs and transfer learning to distinguish NORMAL (no pneumonia findings) from PNEUMONIA chest X-rays. Using the Chest X-Ray Images (Pneumonia) dataset, it implemented data validation, exploratory analysis, a baseline CNN, ResNet18 transfer learning, test evaluation, Grad-CAM explainability, and a FastAPI/Streamlit demo end to end. The original test results were 85.7% accuracy, 99.5% recall, and 0.96 ROC-AUC. This work is for educational and academic purposes and must not be used as a clinical diagnostic tool.

1. Problem definition

1.1 Problem

Input: A grayscale chest X-ray, or one converted to RGB, in JPEG/PNG format.

Output: Binary classification as NORMAL or PNEUMONIA, with a probability for each class and a confidence score. This is modeled as an image classification problem in computer vision.

1.2 Why the problem matters

Pneumonia is a common infection of lung tissue and can cause serious complications if treatment is delayed. Chest X-rays are widely used in clinical practice. Automatic image analysis has long been studied to reduce radiologists' workload and provide early warnings.

Deep learning models can learn patterns in these images. Use in healthcare requires extensive clinical validation, regulation, patient safety measures, and expert oversight. This project addresses the problem within an educational and academic setting; predictions do not replace medical decisions.

2. AI method and rationale

2.1 Method selection

The project uses convolutional neural networks (CNNs) with a two-stage modeling strategy:

Stage	Model	Purpose
Baseline	SimpleCNN (custom architecture)	Verify the dataset and pipeline; establish reference performance
Final in the original version	ResNet18 + transfer learning	Learn stronger representations using an ImageNet-pretrained feature extractor

Reasons:

- **Image data:** The problem is pixel-based; CNNs learn spatial features hierarchically.
- **Limited data:** Training a very deep network from scratch is risky; pretrained ResNet18 can support generalization with less data.
- **ResNet architecture:** Residual connections reduce vanishing gradients; related architectures appear widely in medical image classification, including CheXNet research.
- **Binary classification:** A two-class softmax output and weighted cross-entropy address class imbalance.
- **Explainability:** Grad-CAM visualizes regions associated with model decisions to aid interpretation.

EfficientNet-B0 and MobileNetV3-Small are also supported in the configuration; the original final experiments used ResNet18.

2.2 Training strategy

- **Classifier training (8 epochs):** Freeze the ImageNet backbone and train only the final fully connected layer.
- **Fine-tuning (12 epochs):** Update the final ResNet convolutional block (layer4) and classifier together at a lower learning rate.

Optimization used AdamW, ReduceLROnPlateau, and early stopping (patience=5). The best model was selected by validation loss.

3. Dataset

3.1 Source

Chest X-Ray Images (Pneumonia), distributed through Kaggle/Mendeley, contains pediatric chest X-rays. The original report describes these as anteroposterior images.

Source: Kaggle: Chest X-Ray Images (Pneumonia).

Raw data was placed manually under `data/raw/chest_xray/`; there was no automatic download.

3.2 Structure and statistics

```
chest_xray/  
  train/  (NORMAL, PNEUMONIA)  
  val/    (NORMAL, PNEUMONIA)  
  test/   (NORMAL, PNEUMONIA)
```

Split	NORMAL	PNEUMONIA	Total
Train	1,341	3,875	5,216
Val (raw)	8	8	16
Test	234	390	624
Total	1,583	4,273	5,856

- Corrupted images: 0.
- PNEUMONIA accounts for about 73% of the dataset, indicating substantial class imbalance.

3.3 Validation split: original design decision

The raw validation split has only 16 images, which is insufficient for model selection. The raw training folder was therefore split into 85% training / 15% validation with stratification (seed=42). The raw test split was retained.

Manifest	NORMAL	PNEUMONIA	Total
Training	1,140	3,294	4,434
Validation	201	581	782
Test	234	390	624

3.4 Preprocessing and augmentation

- Input size: 256 x 256 pixels. ImageNet mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225].
- Training: horizontal flips (50%) and controlled rotations of +/-7 degrees.
- Validation/test/inference: deterministic resizing and normalization only.

EDA found variable dimensions (mean approximately 1328 x 971 pixels) and widely varying aspect ratios; aggressive augmentation was avoided.

4. Implemented solution

4.1 Software architecture

The project is structured as a modular Python package:

Module	Responsibility
src/data/	Data validation, manifest generation, PyTorch Dataset and DataLoader
src/eda/	Exploratory analysis and summary reports
src/training/	Model definitions and training loop
src/evaluation/	Test metrics and charts
src/inference/	Single-image prediction
src/explainability/	Grad-CAM
app/api/	FastAPI prediction service
app/ui/	Streamlit demo interface

Central configuration: configs/config.yaml. Original quality checks: pytest (34 tests) and Ruff lint. These counts are historical.

4.2 Experimental setup

Parameter	Value
Original final model	ResNet18 (ImageNet pretrained)
Image size / batch size	256 / 24
Classifier / fine-tuning learning rate	0.0003 / 0.00001
Total epochs	20 (8 + 12)
Seed / device	42 / CUDA (GPU)
Weighted loss	Enabled

4.3 Baseline comparison

Model	Best epoch	Val. accuracy	Val. F1
SimpleCNN (baseline)	8	0.9066	0.9362
ResNet18 (transfer learning)	18	0.9795	0.9862

Transfer learning performed substantially better than the baseline on the original validation set.

5. Experiments and results

5.1 Test metrics: original ResNet18 model

Results were measured on 624 images from the original raw test split, which was not used for training in that experiment.

Metric	Value
Accuracy	0.8574 (85.7%)
Precision	0.8168
Recall	0.9949
F1-score	0.8971
ROC-AUC	0.9599

Class	Precision	Recall	F1
PNEUMONIA	0.8168	0.9949	0.8971
NORMAL	0.9866	0.6282	0.7676

5.2 Confusion matrix

Actual / predicted	NORMAL	PNEUMONIA
NORMAL	147 (TN)	87 (FP)
PNEUMONIA	2 (FN)	388 (TP)

- True positive (TP): 388 - correctly detected pneumonia.
- True negative (TN): 147 - correctly detected normal images.
- False positive (FP): 87 - NORMAL images incorrectly classified as PNEUMONIA.
- False negative (FN): 2 - PNEUMONIA images incorrectly classified as NORMAL.

5.3 Interpretation

Strengths: Very high recall (99.5%), with only two missed positive images; high ROC-AUC (0.96); and improved validation performance with transfer learning.

Weaknesses: 87 false positives and 81.7% precision; low NORMAL-class recall (62.8%), with about one-third of normal images missed; and accuracy of 85.7%, which alone can be misleading on an imbalanced dataset.

High recall may be desirable when avoiding missed pneumonia, but the false positive rate requires threshold review, expert validation, and multicenter testing before any clinical use. This system is not a diagnostic tool.

5. Results: explainability and serving

5.4 Grad-CAM explainability

Grad-CAM heatmaps visualize activations in the final convolutional layer for correct and incorrect predictions. The aim is to examine whether the model focuses on lungs or unrelated features such as borders or markings. Grad-CAM is an approximate explanation, not pathology localization or causal evidence.

5.5 Demo and service layer

- FastAPI: /health, /model-info, /predict; image uploads produce JSON predictions.
- Streamlit: image upload, preview, and result display through the API.
- All outputs carry the warning: "This output must not be used for medical diagnosis."

6. Brief assessment

6.1 Project scope

Requirement	Implementation
Real-world problem	Pneumonia classification from chest X-rays
Problem definition and significance	Section 1
AI method and rationale	CNN + transfer learning (Section 2)
Data use	Open dataset, EDA, stratified split (Section 3)
Implementation	PyTorch pipeline, API, UI (Section 4)
Result analysis	Metrics, confusion matrix, Grad-CAM (Section 5)
Source code	src/, app/, scripts/, tests/, README
Report	This document, suitable for Word/PDF export

6.2 Limitations of the original version

- One dataset and population (mostly pediatric); no external validation in the original version.
- Binary classes; other lung pathologies are not distinguished.
- High false positive rate; the original report predates threshold optimization.
- Clinical approval, regulation, and ethics approval are outside the scope.
- Grad-CAM interpretation does not replace an expert radiologist.

6.3 Conclusion

The original project addressed a meaningful, data-supported medical imaging problem end to end using deep learning. ResNet18 transfer learning achieved strong discrimination (ROC-AUC) and very high pneumonia recall on its test, but false positives remained high. The modular, tested code and demo formed an implementable, measurable, and reportable educational project.

References

- Kaggle - Chest X-Ray Images (Pneumonia). Dataset page.
- Mendeley Data - Chest X-Ray Images (Pneumonia). Dataset record, version 2.
- Rajpurkar, P., et al. (2017). CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning. arXiv:1711.05225.
- He, K., et al. (2016). Deep Residual Learning for Image Recognition. arXiv:1512.03385.
- Selvaraju, R. R., et al. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. arXiv:1610.02391.

Appendix: historical run commands

These commands preserve the original report. For the current audited training/selection workflow, follow README.md.

```
# Installation
python -m pip install -e "[dev]"

# Prepare data after placing it manually
python -m src.data.validate_dataset --config configs/config.yaml
python -m src.data.standardize_dataset --config configs/config.yaml

# Original training and evaluation
python -m src.training.train_baseline --config configs/config.yaml
python -m src.training.train --config configs/config.yaml
python -m src.evaluation.evaluate --config configs/config.yaml

# Demo
python -m uvicorn app.api.main:app --host 0.0.0.0 --port 8000
python -m streamlit run app/ui/streamlit_app.py
```

Project root: pneumonia-xray-cnn. See README.md for detailed usage.